Dr.-Ing. Maximilian Raff

M.Sc. Engineering Cybernetics

—— Contacts ——

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– Personal Details –

Date of birth: 29.10.1993 in Herrenberg
Citizenship: Germany
Marital status: married, one child

—— Skills ——

German	■■■■■
English	■■■■□
Matlab/Simulink	■■■■■
Python	■■■□□
Julia	■■■■□
C++	■■□□□
CAD	■■□□□
Latex	■■■■■
Git	■■■■□

—— Hobbies ——

Music: Trombone player in various ensembles; musical and organizational director of a semi-professional jazz band.

Sports: Badminton instructor; cycling enthusiast; enjoy dancing.

Stuttgart, January 2026

Work Experience

07/2025 – now
Postdoctoral Researcher at University of Stuttgart, Germany
Institute for Adaptive Mechanical Systems (IAMS)
Located at Max-Planck-Institute for Intelligent-Systems
- Research on co-design of robotic systems, with a focus on hardware development.

10/2019 – 07/2025
Research Associate at University of Stuttgart, Germany
Institute for Adaptive Mechanical Systems (since 04/2025)
Institute for Nonlinear Mechanics (until 03/2025)
- Conducted research on dynamics and control of legged systems, leading to multiple collaborations, publications, and conference presentations.
- Authored and secured two DFG grant proposals.
- Supervised and mentored over 20 Bachelor's and Master's students, guiding them through scientific projects.
- Developed and delivered lecture content, actively contributing to teaching and examination processes.
- Supported the launch of IAMS robotics institute, overseeing IT and recruiting secretary and PhD candidates.

Internships & Part-Time Positions

06/2018 – 12/2018
Internship at Trumpf GmbH + Co. KG, Ditzingen, Germany
dept. Machine Dynamics
- Modeling/Simulation of laser cutting machine tools
- Development of visualization tools using computational geometry

12/2017 – 05/2018
Research Assistant at NC A&T TECHLAV Center, Greensboro, NC, USA
- Quadcopter trajectory generation and control design

2015 – 2017
Teaching Assistant at University of Stuttgart, Germany
- Institutes for system dynamics (ISYS), systems theory and automatic control (IST) and nonlinear mechanics (INM)

Education

07/2020 – 07/2025
Doctor of Engineering (Dr.-Ing.), University of Stuttgart, Germany *(Grade: summa cum laude)*
Jointly with Graduate School IMPRS-IS
Thesis: Continuation and Optimization of Gaits and Other Non-Smooth Orbits

10/2016 – 10/2019
Master of Science in Engineering Cybernetics, University of Stuttgart, Germany *(Grade: 1.6)*
Thesis: Exploring Soft Contact Models in the Optimization of Gaits
- **01/2019 – 04/2019:** Visiting Scholar at University of Michigan, Ann Arbor, MI, USA
  - Research in legged robotics/locomotion
  - Collaborations: RAM-Lab & Barton Research Group
- **08/2017 – 05/2018:** Study Abroad in North Carolina, USA
  - Exchange student for an academic year attending two universities:
    - NC A&T State University (2 semesters)
    - University of North Carolina at Greensboro (1 semester)

10/2013 – 11/2016
Bachelor of Science in Engineering Cybernetics, University of Stuttgart, Germany *(Grade: 2.2)*
Thesis: Discrete Optimal Control of Inverted Pendulum with Friction

09/2010 – 06/2013
Abitur, Technisches Gymnasium, Gottlieb-Daimler-Schule Sindelfingen, Germany *(Grade: 1.6)*
Major in information technology